Tacit Knowledge Management with Generative AI: Proposal of the GenAI SECI Model

運用生成式 AI 進行隱性知識管理：GenAI SECI 模型之提案

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Keywords（關鍵詞）: *Knowledge Management*（知識管理）; *Tacit Knowledge*（隱性知識）; *SECI Model*（*SECI* 模型）; *Generative AI*（生成式 *AI*）

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【術語使用說明】
本譯文保留部分原文術語以忠實呈現作者定義：
• **Gen-Ba**（現場）：作者自定義日文術語，指知識被創造、共享與運用的工作場域，全文保留英文原詞並以「（現場）」補充說明。
• **Latent Knowledge**（半顯性知識）：介於顯性與隱性之間、可片段性表達的知識，非一般中文「潛在」之意，故譯為「半顯性知識」以示區別。
• **knowledge fragments**（知識片段）：作者自創術語，指部分性、片段性積累於網路空間的知識單元。
• **Digital Fragmented Knowledge**（數位片段知識）：存在於網路空間的顯性與半顯性知識集合體，非指知識被破壞，而指其片段性的本質。
• **symbol grounding**（符號落地）：AI / 認知科學術語，指符號與現實意義的連結過程，本譯文採「符號落地」。

Abstract | 摘要

English Abstract

The emergence of generative AI is bringing about a significant transformation in knowledge management. Generative AI has the potential to address the limitations of conventional knowledge management systems, and it is increasingly being deployed in real-world settings with promising results. Related research is also expanding rapidly. However, much of this work focuses on research and practice related to the management of explicit knowledge. While fragmentary efforts have been made regarding the management of tacit knowledge using generative AI, the modeling and systematization that handle both tacit and explicit knowledge in an integrated manner remain insufficient. In this paper, we propose the "GenAI SECI" model as an updated version of the knowledge creation process (SECI) model, redesigned to leverage the capabilities of generative AI. A defining feature of the "GenAI SECI" model is the introduction of "Digital Fragmented Knowledge", a new concept that integrates explicit and tacit knowledge within cyberspace. Furthermore, a concrete system architecture for the proposed model is presented, along with a comparison with prior research models that share a similar problem awareness and objectives.

繁體中文摘要

生成式 AI 的興起正為知識管理領域帶來重大轉變。生成式 AI 具備克服傳統知識管理系統限制的潛力，其在實際場景中的應用案例日益增加，成果也頗為可期。相關研究更正以迅猛態勢持續擴展。然而，目前大多數研究仍聚焦於顯性知識（Explicit Knowledge）的管理。儘管已有若干零散性嘗試將生成式 AI 應用於隱性知識（Tacit Knowledge）的管理，但能夠整合處理顯性與隱性知識的系統化建模工作至今仍不充分。本文提出「GenAI SECI」模型，作為知識創造流程（SECI）模型的更新版本，重新設計以充分發揮生成式 AI 的能力。「GenAI SECI」模型的核心特色在於引入「數位片段知識」（Digital Fragmented Knowledge）這一新概念，用以整合存在於網路空間中的顯性知識與隱性知識。此外，本文亦呈現基於所提模型的具體系統架構，並與具有類似問題意識與目標的既有研究模型進行比較。

Keywords（關鍵詞）：Knowledge Management（知識管理）；Tacit Knowledge（隱性知識）；SECI Model（SECI 模型）；Generative AI（生成式 AI）

1. Introduction | 引言

In recent years, the latest AI technologies, including generative AI, have been bringing about major industrial and social transformations — a phenomenon referred to as AI Transformation (AX). For example, business support through conversational AI agents powered by Large Language Models (LLMs), and AI-assisted coding tools such as GitHub Copilot, have the potential not merely to improve operational efficiency, but to fundamentally reshape industrial structures, including the nature of work itself and the way people work.

近年來，以生成式 AI 為代表的最新 AI 技術正引發重大的產業與社會變革，此現象被稱為「AI 轉型」（AI Transformation，AX）。舉例而言，以大型語言模型（LLMs）為核心的對話式 AI 代理人所提供的商業支援，以及 GitHub Copilot 等 AI 輔助程式設計工具，不僅有望改善業務效率，更可能從根本上重塑產業結構，包括工作本質與工作方式本身。

Generative AI is also driving significant change in the field of knowledge management. Conventional knowledge management systems have had several major challenges, and even when organizations built and deployed such systems and accumulated knowledge within them, the systems themselves often could not be put to effective use. Specifically, organizing and registering knowledge into a system required considerable effort, and users frequently struggled to retrieve the information they needed through search. Generative AI, however, holds the potential to resolve these challenges (O'Leary, 2024), and cases of generative AI being applied to knowledge management are indeed on the rise.

生成式 AI 亦在知識管理領域引發顯著變革。傳統知識管理系統存在若干重大挑戰：即便組織已建立並部署了相關系統、在其中累積知識，這些系統往往仍難以被有效運用。具體而言，將知識整理並登錄至系統需要耗費大量心力，使用者也常常難以透過搜尋找到所需資訊。然而，生成式 AI 具有解決這些挑戰的潛力（O'Leary, 2024），實際將生成式 AI 應用於知識管理的案例也確實日益增多。

Nevertheless, the application of generative AI to knowledge management has, until recently, been largely confined to explicit knowledge. At the same time, efforts to extract and leverage tacit knowledge from the workplace (referred to as "Gen-Ba" in Japan) using generative AI have been growing in recent years — a trend that is particularly active in Japan. This is due to the following reasons. Workplace tacit knowledge has long been a source of strength and competitive advantage for Japanese companies. However, as Japan faces an accelerating demographic shift toward an aging society with a declining birthrate, there is an increasingly urgent need to efficiently and effectively transfer workplace tacit knowledge from veteran employees to younger employees before the former retire. In practice, pilot efforts to leverage generative AI for the transfer of workplace knowledge have already begun in leading-edge companies in Japan.

然而，生成式 AI 在知識管理領域的應用，迄今大多侷限於顯性知識的範疇。與此同時，近年來透過生成式 AI 從職場（日本稱為 Gen-Ba（現場））萃取並運用隱性知識的努力正持續增加，此趨勢在日本尤為活躍。原因如下：Gen-Ba（現場）的隱性知識長期以來是日本企業的核心競爭力所在。然而，面對少子高齡化加速的現實，如何在資深員工退休前，有效率地將其職場隱性知識傳承給年輕員工，已成為日益迫切的課題。實際上，日本部分領先企業已開始試行運用生成式 AI 進行職場知識傳承的先導實驗。

While such efforts to extract and leverage workplace tacit knowledge through generative AI are underway, systematic modeling of these efforts is still insufficient. In this paper, we propose the "GenAI SECI" model — a version of the knowledge creation process (SECI) model updated on the premise of generative AI utilization. We also present a concrete system architecture "Digital Knowledge Twin" based on the GenAI SECI model.

儘管透過生成式 AI 萃取並運用職場隱性知識的實踐正在推進，對這些努力的系統性建模工作仍然不足。為此，本文提出「GenAI SECI」模型，即以生成式 AI 應用為前提、對知識創造流程（SECI）模型進行更新的版本。本文亦呈現基於 GenAI SECI 模型的具體系統架構「數位知識孿生」（Digital Knowledge Twin）。

The remainder of this paper is organized as follows. Section 2 presents a review of prior research. Section 3 proposes a knowledge management model that incorporates the use of generative AI, then Section 4 illustrates a concrete system architecture based on the proposed model. Section 5 clarifies the characteristics of the proposed model through comparison with prior studies, and Section 6 offers concluding remarks.

本文結構如下：第 2 節呈現既有研究回顧；第 3 節提出結合生成式 AI 應用的知識管理模型；第 4 節說明基於所提模型的具體系統架構；第 5 節透過與既有研究的比較，闡明所提模型的特色；第 6 節為結語。

2. Literature Review | 文獻回顧

Among knowledge management models, the SECI model proposed by Nonaka & Takeuchi (1995) is widely known. This model divides knowledge into explicit knowledge and tacit knowledge, and posits that organizational knowledge is created through repeated mutual conversion between the two. The knowledge creation mechanism described by this model effectively explained the competitive advantage of Japanese companies at the time. The SECI model consists of four modes of knowledge conversion:

在知識管理模型中，野中郁次郎與竹內弘高（Nonaka & Takeuchi, 1995）所提出的 SECI 模型廣為人知。該模型將知識區分為顯性知識與隱性知識，並主張組織知識是透過兩者之間的反覆相互轉換而得以創造，其所描述的知識創造機制，有效解釋了當時日本企業的競爭優勢。SECI 模型包含四種知識轉換模式：

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Tacit Knowledge ──[Socialization 社會化]──→ Tacit Knowledge ↑ ↓ [Externalization 外化]
[Internalization 內化] ↑ ↓ Explicit Knowledge ←──[Combination 結合]── Explicit Knowledge
┌─────────────────────────────────────────┐ │ SECI 四種知識轉換模式： │ │
Socialization（社會化）：隱性 → 隱性 │ │ Externalization（外化）：隱性 → 顯性 │ │ Combination
（結合）：顯性 → 顯性 │ │ Internalization（內化）：顯性 → 隱性 │
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Figure 1: *SECI Model. (Adapted from Nonaka & Takeuchi (1995))*

【圖 1】**SECI** 模型（改編自 **Nonaka & Takeuchi, 1995**）

• Socialization (Tacit → Tacit): The process by which tacit knowledge is shared within an organization as tacit knowledge through direct experience sharing, observation, and imitation.

• 社會化（隱性→隱性）：透過直接經驗分享、觀察與模仿，將隱性知識作為隱性知識在組織內部加以共享的過程。

• Externalization (Tacit → Explicit): The process of converting tacit knowledge into explicit forms such as concepts, metaphors, models, and documents, making experiential knowledge accessible to the entire organization.

• 外化（隱性→顯性）：將隱性知識轉換為概念、比喻、模型、文件等顯性形式，使經驗性知識得以在整個組織中流通的過程。

• Combination (Explicit → Explicit): The process of generating new explicit knowledge by combining, categorizing, and integrating multiple forms of explicit knowledge.

• 結合（顯性→顯性）：透過組合、分類與整合多種顯性知識，產生新顯性知識的過程。

• Internalization (Explicit → Tacit): The process of absorbing explicit knowledge through practice and experience and embodying it as an individual's tacit knowledge.

• 內化（顯性→隱性）：透過實踐與經驗吸收顯性知識，並將其體現為個人隱性知識的過程。

Because the SECI model is a versatile and easy-to-use framework, various studies based on the SECI model have been conducted in relation to knowledge management utilizing AI including generative AI. These fall into two categories: (1) studies describing the role of AI and generative AI within the four knowledge conversion modes of the SECI model, and (2) studies proposing new models based on the SECI model.

由於 SECI 模型是一個通用性高且易於使用的框架，以其為基礎探討 AI（含生成式 AI）知識管理應用之研究已陸續展開，並可分為兩類：（1）描述 AI 及生成式 AI 在 SECI 四種知識轉換模式中所扮演之角色的研究；（2）以 SECI 模型為基礎提出新模型的研究。

Studies in category (1) include Kai, Purba & Al-Hosaini (2024), Shen & Lin (2024), He & Burger-Helmchen (2025), and Sumbal & Amber (2025). Kai & Purba & Al-Hosaini (2024) is a systematic review of the relationship between AI and knowledge management, pointing out that "AI-driven systems can automate and enhance the processes of externalization and internalization by making tacit and explicit knowledge more accessible and actionable," but does not address generative AI in detail. He & Burger-Helmchen (2025) organizes and discusses how AI can facilitate and accelerate each of the SECI processes, concluding that the most effective use of AI is to complement and augment rather than replace human capabilities. Shen & Lin (2024) propose an AI-assisted Experience-Based Knowledge Management System (EBKMS), explaining each function of the system in correspondence with each process of the SECI model. Sumbal & Amber (2025) explain how ChatGPT can be used in each process of the SECI model.

第（1）類研究包括 Kai, Purba & Al-Hosaini（2024）、Shen & Lin（2024）、He & Burger-Helmchen（2025）以及 Sumbal & Amber（2025）。其中，Kai 等人（2024）是一篇探討 AI 與知識管理關係的系統性回顧，指出「AI 驅動系統可透過提升隱性與顯性知識的可及性與可操作性，自動化並強化外化與內化的流程」，但未對生成式 AI 進行深入討論。He & Burger-Helmchen（2025）整理並探討了 AI 如何促進和加速 SECI 各流程，認為 AI 最有效的用途在於補充與擴增人類能力，而非取代之。Shen & Lin（2024）提出一套 AI 輔助的「基於經驗之知識管理系統」（EBKMS），並對應 SECI 模型各流程說明系統的各項功能。Sumbal & Amber（2025）則說明 ChatGPT 如何應用於 SECI 模型的各個流程。

Studies in category (2) take the position that the SECI model cannot adequately represent knowledge management utilizing generative AI; specifically, these include Böhm & Durst (2026) and Kirchner & Scarso (2026). Böhm & Durst (2026) propose GRAI (Generative Receptive Artificial Intelligence) as a useful framework for describing the new role of machines (generative AI) in the knowledge creation process. GRAI treats machines (generative AI) as new actors and expands the four interactions of the SECI model by adding human–machine (generative AI) dimensions, extending them to eight interactions. Kirchner & Scarso (2026) also treat generative AI as a new actor, and additionally introduce "Artificial Knowledge" — a new type of knowledge generated by generative AI alongside explicit and tacit knowledge — proposing the AKI (Artificial Knowledge Integration) model. All of these are very recent papers published in 2026, and there remains considerable room for further investigation in this field.

第（2）類研究認為 SECI 模型無法充分涵蓋生成式 AI 的知識管理，具體包括 Böhm & Durst（2026）與 Kirchner & Scarso（2026）。Böhm & Durst（2026）提出 GRAI（Generative Receptive Artificial Intelligence）框架，將機器（生成式 AI）視為新行為者，在 SECI 模型的四種互動基礎上增加人機維度，擴展為八種互動。Kirchner & Scarso（2026）同樣將生成式 AI 視為新行為者，並在顯性與隱性知識之外引入「人工知識」（Artificial Knowledge）這一新知識類型，提出 AKI（Artificial Knowledge Integration）模型。以上均為 2026 年發表的最新論文，本領域仍有相當大的深入研究空間。

In this study, while taking the position that a new model is needed for knowledge management utilizing generative AI, we propose a new model based on the SECI model (the GenAI SECI model) that differs from Böhm & Durst (2026) and Kirchner & Scarso (2026) in treating generative AI not as a new actor but strictly as an auxiliary means. What is unique and novel about the GenAI SECI model is that it introduces "Digital Fragmented Knowledge" as a new type of knowledge in cyberspace, and utilizes generative AI for the externalization and internalization of this knowledge.

本文同樣認為生成式 AI 知識管理領域需要新模型，但所提出的 GenAI SECI 模型與 Böhm & Durst（2026）及 Kirchner & Scarso（2026）的根本差異在於：生成式 AI 被定位為輔助工具，而非新的行為者。GenAI SECI 模型的獨特性與創新性在於引入「數位片段知識」（Digital Fragmented Knowledge）作為網路空間中的新知識類型，並運用生成式 AI 進行此類知識的外化與內化。

3. Knowledge Management Model Utilizing Generative AI｜運用生成式 AI 的知識管理模型

In this section, we propose a knowledge management model utilizing generative AI (GenAI SECI model). First, we clarify the knowledge to be managed here — specifically, workplace knowledge ("Gen-Ba" Knowledge (Uchihira et al., 2023)). In Japanese, "Gen" means "actual and physical field," and "Ba" means "knowledge creating space," which was introduced by Nonaka & Takeuchi (1995). Gen-Ba knowledge refers to knowledge that is created, shared, and utilized in the workplace. Gen-Ba includes various settings such as manufacturing sites, maintenance and inspection sites, medical and nursing care settings, agricultural sites, and sales and customer service environments. Furthermore, knowledge here is defined as justified belief that serves as the basis for human action. Here, "justified" means "something that one has genuinely accepted and internalized for oneself." In this paper, Gen-Ba knowledge is classified into three layers: explicit knowledge, latent knowledge, and tacit knowledge (Figure 2). The three layers of knowledge are continuous and do not have clear boundaries between them.

本節提出運用生成式 AI 的知識管理模型（GenAI SECI 模型）。首先，本文釐清所欲管理的知識類型，即 Gen-Ba（現場）知識（Uchihira et al., 2023）。日文中「Gen（現）」意指「實際存在的物理場域」，「Ba（場）」則意指野中郁次郎與竹內弘高（1995）所提出的「知識創造空間」。Gen-Ba 知識是指在工作場域中被創造、共享與運用的知識，涵蓋製造現場、維修保養現場、醫療護理環境、農業現場，以及業務與客戶服務環境等多種情境。此外，本文將知識定義為作為人類行動基礎的「有根據的信念」，其中「有根據」意指「確實為自己所接受並內化的事物」。本文將 Gen-Ba 知識分為三個層次：顯性知識、半顯性知識（Latent Knowledge）與隱性知識（圖 2），三層次之間連續而無明確界限。

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△ Explicit Knowledge（顯性知識） ▽△ Latent Knowledge（半顯性知識） ▽▽△ Tacit Knowledge, Narrow Sense（隱性知識，狹義） 金字塔結構（由上至下，範圍漸廣）：
┌────────────────────────────┐ │ 頂層：Explicit Knowledge（顯性知識） │ │ 可有意識地認知，能完整、系統性地編碼 │ ├────────────────────────────┤ │ 中層：Latent Knowledge（半顯性知識） │ │ 平時不自覺，在 Gen-Ba 或被詢問時 │ │ 可部分、片段性地加以表達 │ ├────────────────────────────┤ │ 底層：Tacit Knowledge, Narrow Sense │ │ （隱性知識，狹義） │ │ 無意識存在，原則上無法以語言表達 │
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Figure 2: Continuous Gen-Ba Knowledge Layers (Uchihira et al., 2023)

【圖 2】連續性的 **Gen-Ba**（現場）知識層次（**Uchihira et al., 2023**）

• Explicit Knowledge: Knowledge that is consciously recognized and can be codified (manualized) in a complete and systematic manner.

• 顯性知識（Explicit Knowledge）：能夠被有意識地認知，且可以完整、系統性地編碼（手冊化）的知識。

• Latent Knowledge: Knowledge that is not ordinarily conscious, but can be partially and fragmentarily expressed as code (text, image, video, etc.) when one is in the Gen-Ba or when asked by others.

• 半顯性知識（Latent Knowledge）：平時不自覺察覺，但在身處 Gen-Ba（現場）或被他人詢問時，可部分性、片段性地以編碼（文字、圖像、影片等）加以表達的知識。

• Tacit Knowledge (Narrow Sense): Knowledge that exists unconsciously and cannot in principle be expressed in code (verbalized). A representative example of tacit knowledge is embodied knowledge such as the skilled techniques of craftspeople.

• 隱性知識（Tacit Knowledge，狹義）：無意識地存在，原則上無法以編碼（語言）加以表達的知識。代表性例子是工匠熟練技藝等身體化知識。

The tacit knowledge of the SECI model encompasses both latent knowledge and tacit knowledge (narrow sense), and is referred to here as tacit knowledge (broad sense). Collins (2010) classified tacit knowledge into "Relational Tacit Knowledge", "Somatic Tacit Knowledge", and "Collective Tacit Knowledge". Somatic Tacit Knowledge corresponds broadly to Tacit Knowledge (Narrow Sense), while Relational Tacit Knowledge corresponds to Latent Knowledge. Regarding Collective Tacit Knowledge, those elements that can be fragmentarily codified are included in Latent Knowledge, while the remainder are classified as Tacit Knowledge (Narrow Sense). It is our view that much of Collective Tacit Knowledge can in fact be partially codified. With respect to Somatic Tacit Knowledge as well, those elements that can be fragmentarily codified may reasonably be included in Latent Knowledge. Here, codification is defined to include meaningful text, images, video, and similar forms of expression.

SECI 模型中的隱性知識同時涵蓋半顯性知識（Latent Knowledge）與隱性知識（狹義），在此合稱為隱性知識（廣義）。Collins（2010）將隱性知識分類為「關係性隱性知識」、「身體性隱性知識」與「集體性隱性知識」。其中，身體性隱性知識大致對應本文的隱性知識（狹義），關係性隱性知識則對應半顯性知識。就集體性隱性知識而言，可片段性編碼的部分納入半顯性知識，其餘則歸入隱性知識（狹義）。本文認為，集體性隱性知識中有相當大比例事實上可以部分性地加以編碼，身體性隱性知識中可片段性編碼的部分亦然。此處的「編碼」定義包含有意義的文字、圖像、影片等表達形式。

Furthermore, the proposed model introduces "Digital Fragmented Knowledge". Explicit Knowledge, Latent Knowledge, and Tacit Knowledge (Narrow Sense) are classifications of knowledge from a human perspective. Digital Fragmented Knowledge is defined as knowledge in cyberspace — specifically, Explicit Knowledge and Latent Knowledge that has been accumulated in cyberspace in some digital form. For example, records of what people in the Gen-Ba have sensed or thought, documented in language or photographs, do not constitute systematic knowledge like a manual, but can be accumulated in cyberspace as partial and fragmentary knowledge (referred to as "knowledge fragments"). While such partial and fragmentary knowledge has traditionally been difficult to handle within knowledge management, generative AI has now made it possible to utilize them. Figure 3 shows the proposed GenAI SECI model.

此外，所提模型引入「數位片段知識」（Digital Fragmented Knowledge）這一概念。顯性知識、半顯性知識與隱性知識（狹義）是從人的視角對知識進行的分類，而數位片段知識則被定義為網路空間中的知識，具體而言，是以某種數位形式積累於網路空間中的顯性知識與半顯性知識的集合體。舉例而言，Gen-Ba（現場）工作者的感知或想法以語言或照片記錄下來的內容，並不構成像手冊那樣的系統性知識，但可以作為部分性、片段性的知識（稱為「知識片段」，knowledge fragments）積累於網路空間中。此類部分性、片段性知識傳統上在知識管理中難以加以運用，但生成式 AI 的出現使其得以被利用。圖 3 呈現所提出的 GenAI SECI 模型。

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┌──────────────────────────────────────────────────────────────┐ │ [Cyber Space 網路空間] │
│ ┌──────────────────────────────────────────────┐ │ │ │ Digital Fragmented Knowledge │ ←──
Combination │ │ 數位片段知識 │ ──→ （知識整合） │ │ (Explicit and Latent Knowledge │ │ │ │
顯性知識與半顯性知識) │ │ │ └──────────────────────────────────────────┘ │ │ ↑
Externalization（外化） ↓ Internalization（內化） │
├──────────────────────────────────────────────────────────────┤ │ [Physical Space 物理空間]
│ │ Gen-Ba Knowledge ←─[Socialization 社會化]── Gen-Ba Knowledge │ （Gen-Ba 知識） （Gen-
Ba 知識） └──────────────────────────────────────────────────────────────┘
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Figure 3: GenAI SECI Model

【圖 3】**GenAI SECI** 模型

In this GenAI SECI model, each process is defined as follows.

在此 GenAI SECI 模型中，各流程定義如下。

• Socialization: Same as socialization in the SECI model. The process by which Gen-Ba knowledge is shared within an organization, through direct experience sharing, observation, and imitation.

• 社會化：與 SECI 模型中的社會化相同。透過直接經驗分享、觀察與模仿，將 Gen-Ba 知識在組織內部加以共享的過程。

• Externalization: The process of codifying Gen-Ba knowledge — even partially and fragmentarily — as "knowledge fragments" and accumulating them in cyberspace.

• 外化：將 Gen-Ba 知識，即便是部分性、片段性地，編碼為「知識片段」（knowledge fragments）並積累於網路空間的過程。

• Combination: The process of organizing the "knowledge fragments" accumulated in cyberspace into a form suitable for internalization.

• 結合：將積累於網路空間的「知識片段」整理成適合內化形式的過程。

• Internalization: The process of referencing the "knowledge fragments" accumulated and organized in cyberspace to amplify human Gen-Ba knowledge.

• 內化：參照積累並整理於網路空間的「知識片段」，以擴增人類 Gen-Ba 知識的過程。

The roles of generative AI in externalization, combination, and internalization of the GenAI SECI model are described below.

以下說明生成式 AI 在 GenAI SECI 模型的外化、結合與內化各流程中所扮演的角色。

• Externalization: When codifying Gen-Ba knowledge distributed across various media (text, sensor data, images, video, etc.) as "knowledge fragments," generative AI aggregates them into meaningful units.

• 外化：在將散布於各種媒介（文字、感測器數據、圖像、影片等）的 Gen-Ba 知識編碼為「知識片段」時，生成式 AI 將其彙整為有意義的單元。

• Combination: Generative AI organizes and stores "knowledge fragments" in a structured format such as a knowledge graph.

• 結合：生成式 AI 以知識圖譜等結構化格式對「知識片段」進行整理與儲存。

• Internalization: Generative AI selects and presents "knowledge fragments" that are effective for amplifying human Gen-Ba knowledge.

• 內化：生成式 AI 選取並呈現對擴增人類 Gen-Ba 知識有效的「知識片段」。

The distinctive feature of this model is that, rather than externalizing tacit knowledge into complete explicit knowledge and then using that explicit knowledge for internalization as in the SECI model, it leverages generative AI to amplify Gen-Ba knowledge through internalization using incomplete knowledge fragments, without requiring full externalization into explicit knowledge. A comparison with the GRAI framework and the AKI model from prior studies is provided in the Discussion section.

此模型的鮮明特色在於：有別於 SECI 模型中將隱性知識完整外化為顯性知識、再以該顯性知識進行內化的做法，本模型無需將知識完整外化為顯性形式，而是運用生成式 AI，透過不完整的知識片段進行內化，以擴增 Gen-Ba 知識。與既有研究 GRAI 框架和 AKI 模型的比較將在討論節呈現。

4. Digital Knowledge Twin System｜數位知識孿生系統

In this section, as a concrete system architecture based on the GenAI SECI model, we introduce the Digital Knowledge Twin System (Uchihira et al., 2025) shown in Figure 4, which consists of three processes (A), (B) and (C). This system is currently under development.

本節作為基於 GenAI SECI 模型的具體系統架構，介紹圖 4 所示的數位知識孿生系統（Uchihira et al., 2025），該系統由三個流程（A）、（B）與（C）構成，目前仍在開發中。

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┌──────────────────────────────────────────────────────────────────┐ │ [Cyber Knowledge
Space 網路知識空間] │ │ (B) Loose Gen-Ba Knowledge Combination（鬆散整理的 Gen-Ba 知識） │ │
┌──────────────────────────────────────────────┐ │ │ │ Operation → Operation → Operation
→ Operation│ │ │ └──────────────────────────────────────────────┘ │ │ ↕ Automatic
Recommendation（自動推薦）↕ │ │ Gen-Ba Knowledge Fragment（Gen-Ba 知識片段） │ │
Collaborative Symbol Grounding Workshop（協作符號落地工作坊） │
└──────────────────────────────────────────────────────────────────┘ ↑(A)
Externalization（外化） (C) Internalization（內化）↓
┌──────────────────────────────────────────────────────────────────┐ │ [Physical
Knowledge Space 物理知識空間] │ │ Semi-automatic Aggregation（半自動彙整） │ │ Speech │
Photo │ Movie │ Physical Sensor Data │ │ Physical & Human Sensor ────────────────────
Gen-Ba Knowledge │ └──────────────────────────────────────────────────────────────────┘
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Figure 4: *Digital Knowledge Twin System (Uchihira et al., 2025)*

【圖 4】數位知識孿生系統（**Uchihira et al., 2025**）

(A) Semi-automatic Aggregation in Externalization｜外化中的半自動彙整

We have extended the Smart Voice Messaging System, developed since 2010 as a human sensor to collect the awareness of skilled workers through voice messages and photos (Uchihira, 2013). This extension allows for the multifaceted recording and accumulation of voice messages, photos, and physical sensor data in the Gen-Ba. In our previous research, the aggregation of voice messages, photos, and physical sensor data (Gen-Ba knowledge fragment) was manually performed by humans, which was a significant burden. This system introduces a semi-automated aggregation technology that leverages generative AI to interactively and quickly consolidate Gen-Ba knowledge fragments that can be collaboratively symbol-grounded in workshops (C).

本研究對自 2010 年起作為人體感測器開發、透過語音訊息與照片收集熟練工作者感知的「智慧語音訊息系統」（Uchihira, 2013）進行了擴展，使其能夠在 Gen-Ba（現場）對語音訊息、照片及物理感測器數據進行多元記錄與積累。在先前的研究中，語音訊息、照片與物理感測器數據（Gen-Ba 知識片段）的彙整作業需由人工執行，負擔相當沉重。本系統引入運用生成式 AI 的半自動彙整技術，能夠以互動方式快速整合可在工作坊（C）進行協作符號落地的 Gen-Ba 知識片段。

(B) Loose Gen-Ba Knowledge Combination｜Gen-Ba 知識的鬆散整理連結

There is a need for technology that loosely links Gen-Ba knowledge fragments, which are captured in the Gen-Ba through (A), with documents such as manuals of field operations. This loose structuring of Gen-Ba knowledge is necessary for utilization in internalization workshops (C). Here, we extend the procedure-based and purpose-based knowledge graph (Ijuin et al., 2023) (Inoue et al., 2023), which has been developed and proven effective in caregiving and maintenance inspection fields. By leveraging generative AI technology, we are developing an interactive method to link field work processes and objects with Gen-Ba knowledge fragments using the knowledge graph. Loosely structured Gen-Ba knowledge fragments will be accumulated in the digital fragmented knowledge database.

有一種技術需求是：將透過（A）在 Gen-Ba（現場）捕捉的知識片段，與現場作業手冊等文件進行鬆散連結。這種對 Gen-Ba 知識的鬆散整理，是在內化工作坊（C）中加以運用的必要條件。本研究在護理及維修保養領域中已開發並驗證有效的「基於程序的知識圖譜」與「基於目的的知識圖譜」（Ijuin et al., 2023；Inoue et al., 2023）基礎上進行擴展，運用生成式 AI 技術，開發以知識圖譜將現場作業流程與對象和 Gen-Ba 知識片段相互連結的互動式方法。經鬆散整理的 Gen-Ba 知識片段將積累於數位片段知識資料庫中。

(C) Automatic Recommendation in Internalization｜內化中的自動推薦

The system supports efficient and effective collaborative symbol grounding (internalization) in workshops involving both skilled and unskilled participants, utilizing loosely structured digital fragmented knowledge and generative AI. Specifically, we develop (1) automatic classification technology for Gen-Ba knowledge to make workshops more efficient (Ogawa et al., 2024), and (2) recommendation technology for Gen-Ba knowledge to make workshops more effective and stimulate discussions (Ogawa et al., 2025).

本系統運用鬆散整理的數位片段知識與生成式 AI，支援由熟練者與非熟練者共同參與的工作坊中，高效且有效的協作符號落地（內化）。具體而言，開發：（1）提升工作坊效率的 Gen-Ba 知識自動分類技術（Ogawa et al., 2024），以及（2）提升工作坊效果並促進討論的 Gen-Ba 知識推薦技術（Ogawa et al., 2025）。

While this Digital Knowledge Twin System has been partially developed, the full integration and evaluation of generative AI remain tasks for the future.

雖然數位知識孿生系統已部分開發完成，但生成式 AI 的完整整合與評估仍有待未來繼續推進。

5. Discussion | 討論

Knowledge management systems that leveraged AI in the twentieth century required knowledge to be described in a fully formalized manner — as logical expressions or IF-THEN rules — at the time of registration, which demanded enormous effort in knowledge preparation. Furthermore, these systems faced the critical limitation that the knowledge needed at the time of use could not always be successfully retrieved, ultimately resulting in systems that were rarely put to practical use. In contrast, the essential characteristic of generative AI lies in its ability to make use of even incomplete or fragmented knowledge that has not been fully formalized.

二十世紀運用 AI 的知識管理系統要求在登錄時以完全形式化的方式描述知識，如邏輯表達式或 IF-THEN 規則，這需要耗費大量的知識準備工作。此外，這些系統還面臨一個關鍵限制：在使用時所需的知識並不總能成功檢索到，最終導致這些系統幾乎無法投入實際使用。相比之下，生成式 AI 的本質特徵在於其能夠利用尚未完全形式化的不完整或片段性知識。

The GenAI SECI model proposed in this paper is distinguished by its introduction of "Digital Fragmented Knowledge" — a body of partial and fragmented knowledge — as a new concept, while preserving the four knowledge conversion processes of the original SECI model, and by leveraging the capabilities of generative AI to collect, integrate, and utilize such knowledge. The position that existing models require revision to adequately address knowledge management with generative AI is shared with the GRAI framework of Böhm & Durst (2026) and the AKI model of Kirchner & Scarso (2026). However, a fundamental distinction of the proposed model from both is that generative AI is positioned not as a new actor or agent, but strictly as an auxiliary means to support human knowledge creation (Table 1).

本文所提出的 GenAI SECI 模型，在保留原有 SECI 模型四種知識轉換流程的同時，引入「數位片段知識」這一部分性、片段性知識的集合體作為新概念，並運用生成式 AI 的能力對此類知識進行收集、整合與運用，此為本模型的鮮明特色。既有模型需要修訂以充分應對生成式 AI 知識管理之立場，與 Böhm & Durst（2026）的 GRAI 框架及 Kirchner & Scarso（2026）的 AKI 模型相同。然而，所提模型與兩者的根本差異在於：生成式 AI 被定位為輔助工具，而非新的行為者或代理人（表 1）。

The AKI model introduced "Artificial Knowledge" as the type of knowledge handled by generative AI. "Artificial Knowledge," generated by AI in cyberspace and refined through contextual understanding in dialogue with humans, shares certain characteristics with "Digital Fragmented Knowledge" in the present model. Nevertheless, there is an important distinction between the two. Whereas "Artificial Knowledge" is presented to humans in an organized, explicit form, what is extracted from "Digital Fragmented Knowledge" and presented to humans consists of "knowledge fragments" — which may or may not take the form of explicit knowledge — and their internalization must be carried out proactively and reflectively by the human gaining insights from "knowledge fragments".

AKI 模型引入「人工知識」（Artificial Knowledge）作為生成式 AI 所處理的知識類型。「人工知識」由 AI 在網路空間中生成，並透過與人類的對話深化脈絡理解而持續精煉，這與本模型的「數位片段知識」具有某些相似之處。然而，兩者之間存在重要區別：「人工知識」以有組織的顯性形式呈現給人類；而從「數位片段知識」中萃取並呈現給人類的是「知識片段」，其可能或不可能具有顯性知識的形式，且其內化必須由人類主動地、反思性地從「知識片段」中自行獲取洞見來完成。

For knowledge to become practical, action-oriented understanding grounded in the actual workplace, it is essential that humans undergo a process of genuine comprehension — what might be described as knowledge "clicking into place" — by integrating new knowledge with their own experiential understanding. For this reason, the GenAI SECI model places particular emphasis on workshops, as illustrated through the case of the Digital Knowledge Twin System. This internalization process corresponds to "Reflective Observation" in Kolb's (1984) experiential learning model. Moreover, by incorporating not only one's own experiences and self-recorded "knowledge fragments" but also the "knowledge fragments" of colleagues in the same workplace, it is conceived that generative AI-augmented individual reflective observation, or collaborative reflective observation among organizational members through workshops, can be realized.

為使知識成為植根於實際職場的、具有行動導向的實踐性理解，人類必須經歷一個真正理解的過程，即可以描述為知識「到位」（clicking into place）的過程，通過將新知識與自身的經驗性理解融合而實現。為此，GenAI SECI 模型特別強調工作坊的重要性，這在數位知識孿生系統的案例中有所體現。此內化過程對應 Kolb（1984）體驗式學習模型中的「反思性觀察」（Reflective Observation）。此外，透過納入不僅是自身的經驗與自我記錄的「知識片段」，還包括同一職場同事的「知識片段」，可以設想實現由生成式 AI 擴增的個人反思性觀察，或透過工作坊在組織成員間實現的協作性反思觀察。

As prior research has noted, knowledge generated by generative AI is not, in itself, symbol-grounded. It is the human being who performs symbol grounding. The most distinctive and defining feature of the GenAI SECI model is precisely this: the use of generative AI to facilitate and support human-led symbol grounding — that is, the internalization of knowledge by the human subject.

正如既有研究所指出的，由生成式 AI 生成的知識本身並不具備符號落地性（symbol grounding），執行符號落地的是人類。GenAI SECI 模型最具獨特性和決定性的特色正在於此：運用生成式 AI 促進並支援人類主導的符號落地，即人類主體的知識內化。

Table 1. Comparison of GRAI, AKI, and GenAI SECI models | 表 **1. GRAI**、**AKI** 與 **GenAI SECI** 模型之比較

比較項目 **Dimension**	**GRAI model** **Böhm & Durst (2026)**	**AKI model** **Kirchner & Scarso (2026)**	**GenAI SECI model** 本文 **This paper**
Model foundation 模型基礎	Extension of the SECI model. Retains the original four conversion modes while adding an agent layer. SECI 模型的擴展版，保留原有四種轉換模式並增加代理人層。	New model inspired by the SECI model. Preserves the knowledge conversion framework. 以 SECI 模型為靈感的新模型，保留知識轉換框架。	Inherits the four SECI conversion processes. Positioned as an expansion to the existing model. 繼承 SECI 四種轉換流程，定位為對既有模型的擴展。
Role of GenAI 生成式 AI 之角色	New actor 新行為者	New actor 新行為者	Auxiliary means 輔助工具
Types of knowledge 知識類型	Tacit, Explicit Knowledge 隱性知識、顯性知識	Tacit, Explicit, Artificial Knowledge 隱性知識、顯性知識、人工知識	Tacit, Explicit, Digital Fragmented Knowledge 隱性知識、顯性知識、數位片段知識
Approach to internalization 內化取向	Promotes internalization through AI-driven learning support, personalization, and role-play simulations. 透過 AI 驅動的學習支援、個人化與角色扮演模擬促進內化。	Human learning from AI within Mentoring Ba. Humans absorb explicit knowledge provided by AI (Absorption mode). 在 Mentoring Ba 中人類向 AI 學習，以吸收模式吸收 AI 提供的顯性知識。	Corresponds to Reflective Observation in Kolb's (1984) experiential learning model. Emphasizes workshop-based experiential understanding. 對應 Kolb（1984）體驗式學習模型中的反思性觀察，強調工作坊式的體驗性理解。
Symbol grounding 符號落地	No explicit mention. 未明確提及。	Implies that Artificial Knowledge is not symbol-grounded, but offers no concrete countermeasures. 隱含人工知識不具符號落地性，但未提供具體對策。	Explicitly states that symbol grounding is performed by humans. GenAI is positioned as a tool to support this process. 明確指出符號落地由人類執行，生成式 AI 定位為支援此過程的工具。
Model complexity 模型複雜度	Moderate: Conceptually straightforward extension via the addition of an agent layer. 中等：透過增加代理人層進行概念上直觀的擴展。	High: 9 conversion modes and 9 Ba. Loss of parsimony is acknowledged as a limitation. 高：9 種轉換模式與 9 個 Ba，簡潔性喪失被認為是其局限性。	Simple: Retains the original 4 conversion modes. Prioritizes ease of implementation and practical applicability. 簡單：保留原有 4 種轉換模式，優先考量易於實施性與實踐適用性。

6. Conclusion | 結論

This paper addressed a critical challenge in the field of knowledge management: the immaturity of systematic knowledge management models that handle both tacit and explicit knowledge in an integrated manner in the context of generative AI utilization. While generative AI has rapidly expanded the possibilities for managing explicit knowledge, the fragmentary and partial nature of workplace tacit and latent knowledge has remained largely unaddressed by existing theoretical frameworks. To bridge this gap, we proposed the "GenAI SECI model" as an updated version of the SECI model, redesigned around the capabilities of generative AI.

本文致力於解決知識管理領域的一個重要挑戰：在生成式 AI 應用的背景下，能夠整合處理隱性知識與顯性知識的系統性知識管理模型尚不成熟。儘管生成式 AI 已迅速拓展顯性知識管理的可能性，職場隱性知識與半顯性知識（Latent Knowledge）的片段性本質，在很大程度上仍未被現有理論框架充分應對。為彌合這一差距，我們提出以生成式 AI 能力為中心、重新設計的 SECI 模型更新版本「GenAI SECI 模型」。

The central contribution of the GenAI SECI model is the introduction of "Digital Fragmented Knowledge" as a new category of knowledge. Unlike conventional knowledge management systems that required fully formalized, rule-based descriptions, Digital Fragmented Knowledge captures partial and fragmentary field knowledge (knowledge fragments) accumulated in cyberspace. The key insight is that generative AI has made it possible to utilize such incomplete knowledge, which existing knowledge management frameworks previously struggled to handle. In the GenAI SECI model, the four original SECI conversion processes are retained, while generative AI is strategically applied to Externalization, Combination, and Internalization to collect, integrate, and leverage these knowledge fragments.

GenAI SECI 模型的核心貢獻在於引入「數位片段知識」（Digital Fragmented Knowledge）作為新的知識類別。有別於傳統知識管理系統要求完全形式化、基於規則描述的做法，數位片段知識捕捉積累於網路空間中的部分性、片段性 Gen-Ba 知識（知識片段）。核心洞見在於：生成式 AI 使得運用此類不完整知識成為可能，而這正是既有知識管理框架過去難以應對之處。在 GenAI SECI 模型中，原有的四種 SECI 轉換流程得以保留，同時生成式 AI 被策略性地應用於外化、結合與內化，以收集、整合並運用這些知識片段。

As prior research has noted, knowledge generated by generative AI is not, in itself, symbol-grounded. Symbol grounding must be performed by human beings through their own lived experience and reflection. The most defining feature of the GenAI SECI model is that it uses generative AI precisely to facilitate and support this human-led symbol grounding — that is, the internalization of knowledge fragments into genuine tacit understanding. This represents a point of departure from preceding models and constitutes the originality and novelty of the proposed approach.

正如既有研究所指出的，由生成式 AI 生成的知識本身並不具備符號落地性，符號落地必須由人類透過自身的親身經歷與反思來完成。GenAI SECI 模型最具決定性的特色在於：運用生成式 AI 正是為了促進並支援這種人類主導的符號落地，即將知識片段內化為真正的隱性理解。這代表著與既有模型的根本性出發點差異，構成所提方法的原創性與新穎性。

To demonstrate the practical applicability of the proposed model, this paper also presented the Digital Knowledge Twin System, currently under development, showing that the model is not merely conceptual but can be realized as a concrete system architecture.

為展示所提模型的實踐可行性，本文亦呈現目前正在開發中的數位知識孿生系統，說明該模型不僅停留於概念層面，而是可作為具體系統架構加以實現。

At the same time, the GenAI SECI model remains an early-stage proposal, and a number of challenges remain to be addressed. First, empirical validation across diverse workplace settings — including manufacturing, healthcare, agriculture, and maintenance — is needed to assess the model's generalizability and to refine its theoretical foundations. Second, further conceptual development of Digital Fragmented Knowledge is required. Finally, as generative AI capabilities continue to evolve rapidly, it will be essential to continuously reassess the model's assumptions and scope in order to ensure that it remains a relevant and robust framework for knowledge management practice.

與此同時，GenAI SECI 模型仍屬早期階段的提案，尚有若干挑戰有待解決。首先，需要在製造業、醫療保健、農業及維修保養等多元職場環境中進行實證驗證，以評估模型的普適性並精煉其理論基礎。其次，需要對數位片段知識進行進一步的概念發展。最後，隨著生成式 AI 能力持續快速演進，有必要不斷重新審視模型的假設與範疇，以確保其在知識管理實踐中持續保有相關性與穩健性。

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